

CHAPTER 8

Architecture Governance



This chapter is part of the GREEN reading tour. See Figure 1-1 in Chapter 1, Section 1.3, for alternative reading tours.

Architecture governance is more than just a business management tool – it's the backbone that ensures everything runs smoothly. It is best thought of as the leadership compass, guiding the organization with a deep understanding of its structure, clear direction, and the ability to fine-tune processes, all with the aim of driving the organization's strategic goals forward.

At its core, architecture governance boils down to one key mission: to streamline an organization's architectural needs into a coherent set of policies, processes, procedures, and standards. The Architecture Governance Framework makes sure the organization's vision and standards line up with real-world business demands.

8.1. Managing the Architecture

Architecture governance is not a theoretical exercise. It is rooted in the day-to-day realities of business. Enterprise Architects play a key role in deploying and sustaining strategies that keep the business on track.

Architecture governance is a shared responsibility among business leaders, not just the responsibility of a CIO or CTO. Enterprise Architects, domain experts, and other key team members provide essential input. A well-structured Architecture Governance Framework helps organizations reduce costs and risks while speeding up decision-making and project delivery.

The TOGAF Standard suggests that architecture governance is intended to control and manage an organization's architecture and the Architecture Capability (the executive body behind the architecture). To this end, the TOGAF Standard has developed a framework that shows the *possible structure* of architecture governance.

I deliberately say the “possible structure” because the Governance Framework shown in Figure 8-1 serves as a reference point. The TOGAF Standard provides the framework as a handle – a starting point – indicating that it is wise to adapt the framework to the situation within one's own organization. Therefore, the framework shown here is by no means the only correct representation of an Architecture Governance Framework.

According to the TOGAF Standard, architecture governance can be captured in the following Governance Framework (Figure 8-1).

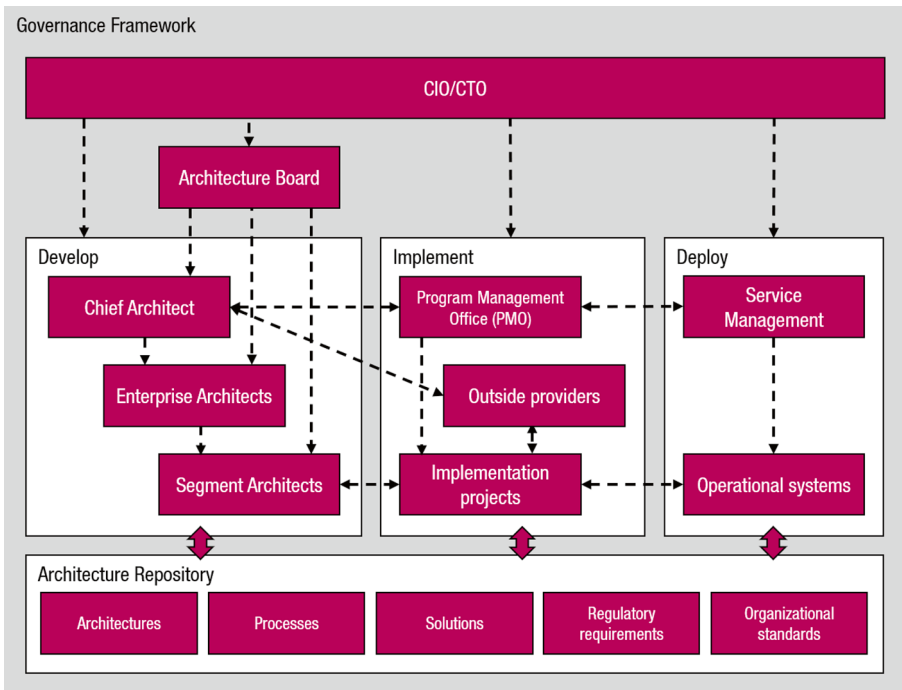


Figure 8-1. *Architecture Governance Framework*

In this Governance Framework, the overall direction is provided by the CIO/CTO. This role is responsible for the direct management of the Architecture Board and the various capabilities responsible for the development, implementation, and deployment of the architecture. These capabilities are performed by various teams within the organization.

The *development* capability (or team) consists of different architecture roles. For this capability, the framework distinguishes the roles of Chief Architect, Enterprise Architect, and Segment Architect (the former Domain Architect). The different roles are explained in more detail in Chapter 10, Section 10.2.

The capability responsible for *implementing* the architecture recognizes the roles of a Program Management Office (PMO), external vendors, and various implementation projects to actually implement the architecture.

The third capability, also governed by the CIO/CTO, is responsible for the actual *deployment* of the architecture. Here the architecture interacts with service management and the various operational systems into which the architecture must fit. Where the architecture will impact existing systems - for example, by requiring modification or possibly even replacement - the connection to the operational systems must be carefully considered.

All capabilities use the Architecture Repository during the architecture initiative. This repository describes the architecture, the processes and solutions to be used, regulatory requirements, and organizational standards. These are all used in the development, implementation, and deployment of the architecture. The Architecture Repository is updated continuously during the initiative to ensure an accurate representation of the developing architecture.

The acronyms CIO and CTO stand for Chief Information Officer and Chief Technology Officer, respectively.

Looking at the first role, that of the CIO (Chief Information Officer), it's clear that this position is almost always part of the IT department. This is odd because information ownership really lies with the business side. Therefore, the CIO, who is responsible for that information, should also be on the business side.

The CTO, on the other hand – given the word “*technology*” in the name – would be perfectly at home in an IT department.

Yet nine out of ten organizations place the CIO in the IT department. The TOGAF Standard also assumes this scenario.

In addition to the existing roles of CIO and CTO, I think the framework would do well to include the role of CEO (Chief Executive Officer) in the Governance Framework.

8.2. Implementing Architecture Governance

Implementing an Architecture Board involves setting up a structured *process* and *governance body* that oversees and guides architectural decisions within an organization. This governance ensures alignment with business goals, standards, and best practices. Phase G (Implementation Governance) of the Architecture Development Method takes the approach described in this section and puts it into practice.

Implementing architecture governance boils down to the following steps. First, the Governance Framework must be established. The necessary roles and responsibilities are defined, and policies and standards are developed. The Governance Framework uses an architecture compliance process to verify that the architecture lives up to the agreements. The architecture review process is implemented, and key performance indicators are established. The entire governance process is then integrated into the Architecture Development Methodology. Next, a Communications Plan is created, and appropriate communication materials are selected or created. The governance processes are automated where possible. Finally, the governance process is subject to continuous improvement.

The steps just described are illustrated in Figure 8-2 and explained in more detail below.

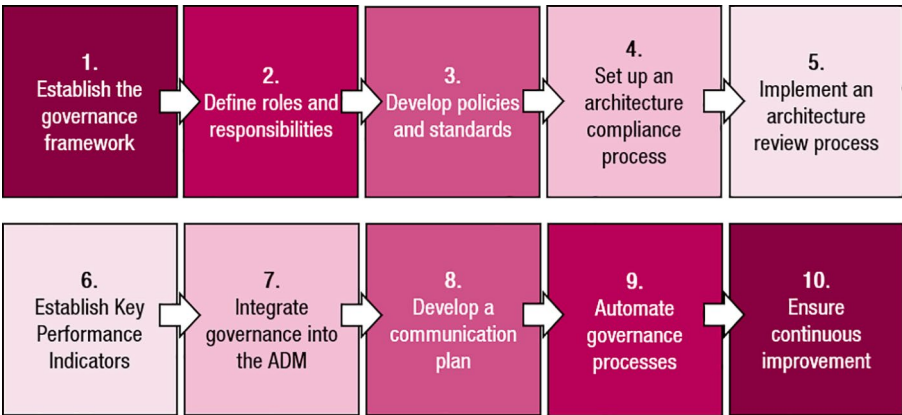


Figure 8-2. *Ten steps for implementing architecture governance*

Implementing architecture governance in a practical and pragmatic way can be achieved by following these steps:

1. Establish the Architecture Governance

Framework: Ensure buy-in from stakeholders by clearly explaining the boundaries and benefits of architecture governance.

- **Kick Off with Stakeholder Engagement:**
Organize a series of workshops or meetings with key stakeholders (e.g., C-level management, IT leadership, business managers) to explain the value of architecture governance. Ensure buy-in by presenting its role in aligning IT with business goals.
- **Create a Governance Charter:** Draft a document outlining the architecture governance scope, objectives, and guiding principles. This document should clearly define

- Which areas of the organization will be governed (business processes, data, IT systems, etc.)
 - What success looks like (alignment with strategic objectives, compliance with standards)
 - **Select Architecture Governance Framework**
Components: Use the TOGAF Standard as a reference to decide which governance processes and structures to implement (approvals, compliance reviews, architecture change management).
 - **Architecture Governance Board Setup:** Identify and invite senior stakeholders and domain architects to form the Architecture Governance Board. Typically, this includes business leaders, IT leaders, Enterprise Architects, and project managers. Define their roles and establish regular governance meetings (e.g., monthly or at key project milestones).
 - **Formalize the Framework:** Document the Governance Framework. This includes governance processes (decision-making, review, and escalation) and the criteria for assessing architecture decisions.
2. **Define Governance Roles and Responsibilities:** Identify, describe, and document necessary roles and responsibilities.
- **Organize Role Definition Workshop:** Conduct a workshop with key stakeholders to define roles (such as Architect, Business Leader, Project

Manager). Identify what each role is accountable for and ensure they understand their governance responsibilities.

- **Create Role Descriptions:** Write detailed role descriptions that clearly state each person’s involvement in architecture governance, including responsibilities for architecture reviews, compliance checks, and approvals.
- **Map Responsibilities:** Use a RACI matrix (see Table 8-1) to map each governance activity to roles. This ensures clarity on who makes decisions and who needs to be informed or consulted.

Table 8-1. *Example RACI Matrix*

Activity ▼	Role	Architect	Business Leader	Project Manager
Establish the Architecture Governance Framework.		A	R	C
Develop governance policies and standards.		R	A	C
Set up an architecture compliance process.		AR	C	I
Implement an architecture review process.		AR	C	I
Establish key performance indicators (KPIs).		R	A	C
Integrate governance into the Architecture Development Method.		AR	C	I
Develop a Communications Plan.		AR	C	C
Automate governance processes.		A	R	R
Ensure continuous improvement.		R	A	A

The RACI matrix shown in Table 8-1 is an example of what such a matrix might look like. It lists the *activities* (or processes) in the left column and shows the *roles* involved in those activities in the top row. The cells of the table contain the manner in which each of the roles is involved in the activities.

For example, when it comes to *establishing the Architecture Governance Framework*, the Architect role is *accountable* for this activity. The Business Leader is *responsible* for the activity, and the Project Manager needs to be *consulted* during the process. The four RACI responsibilities are visualized and explained in Table 8-2.

Table 8-2. *The four RACI responsibilities explained*

Responsibility	Description
Responsible (R)	The person who actually carries out the activity or process. Responsible to get the job done.
Accountable (A)	The person who is ultimately accountable for the activity or process being completed appropriately. Responsible person(s) is/are accountable to this person.
Consulted (C)	People who are not directly involved with carrying out the activity or process, but who are consulted. May be a stakeholder or a subject matter expert.
Informed (I)	Those who receive output from the activity or process or who have a need to stay informed.

A RACI matrix can contain any number of activities and roles. In this particular example, only one Architect role is shown. Which roles are included in the matrix depends on a number of factors, such as the topic for which the matrix is being created.

3. **Develop Governance Policies and Standards:**

Form an interest group to develop policies and standards. Make sure to have them reviewed and published.

- **Form a Policy-Drafting Group:** Select a small team of architects, business leaders, and business representatives to create architecture policies. This group should review existing organizational policies and identify gaps where new governance rules are needed (how architectural changes are approved).
- **Create Draft Policies and Standards:** Start with high-level governance policies that apply to all areas (e.g., *All architecture changes must be reviewed by the governance board*). Next, break these into specific standards, covering technology, processes, and data handling.
- **Review with Stakeholders:** Share draft policies and standards with business leaders and other relevant parties for feedback. Incorporate this feedback into the final documents.
- **Publish and Communicate:** Once finalized, publish the policies and standards on a shared platform (such as the company's intranet). Host training sessions or workshops to explain these to teams involved in architecture projects.

4. **Set Up an Architecture Compliance Process:**

Embed the compliance process in the organization by integrating it in the project lifecycle. Remember to include an exception handling process.

- **Develop Compliance Checklists:** For each architecture domain (Business, Data, Application, and Technology), create checklists that detail compliance requirements. These checklists should align with the governance policies and standards. See the *Further Reading* information box at the end of this section for more information.
 - **Integrate Compliance into Project Lifecycle:** Add compliance checkpoints to key phases of the project lifecycle (during design reviews, before deployment). These checkpoints should mandate sign-off from the Architecture Governance Board.
 - **Compliance Review Meetings:** Schedule regular compliance reviews, either as part of project status meetings or separately. During these reviews verify that projects meet the defined architecture standards. Also, identify and document any exceptions or deviations.
 - **Develop Exception Handling Processes:** When compliance issues arise, have a formalized process for raising exceptions to the governance board. Ensure that a rationale for the deviation is provided and that an action plan for mitigation is agreed upon.
5. **Implement an Architecture Review Process:** Implement a formal review process for architecture documents by developing templates and documenting the outcome of the review meetings.
- **Set Up Review Templates:** Develop standardized templates for architecture reviews. These templates

should include sections for project context, compliance with Architecture Principles, risks, and proposed solutions.

- **Schedule Regular Review Meetings:** Ensure that architecture review sessions are built into the project plan at key phases (design phase, pre-deployment). Send out invitations well in advance and include all relevant stakeholders.
 - **Prepare for Reviews:** Architects or project leads should prepare their review documents in advance. Use the templates to guide their preparation, focusing on key areas such as risk, compliance, and alignment with business objectives.
 - **Review and Document Decisions:** During the review meeting, evaluate each architecture component against established governance standards. Record decisions, actions, and any exceptions granted.
6. **Establish Key Performance Indicators (KPIs):** Identify metrics to measure the performance of the governance activities. Use clearly defined baselines and targets and provide transparent monitoring of the performance.
- **Define KPIs with Stakeholders:** Hold a meeting with key governance stakeholders to identify metrics that will measure the success of governance activities. KPIs could include
 - Percentage of projects compliant with the architecture framework

- Number of governance reviews conducted
 - Number of deviations or exceptions from standards
 - Time to resolve architecture-related issues
- **Set Baselines and Targets:** Once KPIs are defined, set baseline measurements and targets for improvement (e.g., *90% of projects will be architecture-compliant by the end of the year*).
 - **Create Monitoring Dashboards:** Use an architecture management tool or project management software to track KPIs in real time. Ensure that governance board members can access the dashboard to monitor progress. An example of how to monitor progress is shown in Figure 12-54 in Chapter 12, Section 12.10.2.
7. **Integrate Governance into the Architecture Development Method:** Define governance checkpoints and integrate them into the Architecture Development Method phases.
- **Add Governance to Each ADM Phase:** For each phase of the Architecture Development Method – from the Preliminary Phase to Implementation Governance and beyond – define governance checkpoints. For example:
 - **Preliminary Phase:** Ensure Governance Framework setup is complete.
 - **Phase B (Business Architecture):** Perform reviews to validate alignment with business goals.

- **Governance Sign-Off at Key Milestones:** Embed formal governance reviews into project milestones. Projects should not proceed to the next phase without governance board sign-off.
8. **Develop a Communications Plan:** Identify the audience that is impacted by the architecture governance. Select, create, and use appropriate communication materials.
- **Identify Key Audiences:** Create a list of individuals and groups impacted by architecture governance (such as project managers, developers, business stakeholders).
 - **Develop Communication Materials:** Prepare materials such as slide decks, FAQs, and email templates explaining the governance process, its benefits, and their roles. Ensure materials are tailored to different audiences (e.g., technical teams get detailed compliance information; business leaders get strategic benefits).
 - **Host Workshops and Training:** Schedule live workshops or recorded webinars to walk stakeholders through the Architecture Governance Framework, compliance expectations, and how to participate in reviews.

The development of a Communications Plan is further detailed in Chapter 14, Section 14.3.3.

9. **Automate Governance Processes:** If possible, try to automate (parts of) the governance process such as generating reports and review notifications.

- **Select a Governance Tool:** Choose a tool that supports architecture governance, such as Enterprise Architecture management software. These tools can help automate compliance tracking, generate reports, and manage architecture documentation. See Chapter 7, Section 7.7.1, for more information on tool selection.
 - **Set Up Automated Alerts and Reports:** Use the tool to set up automated notifications for governance board members when projects are due for review. Similarly, configure reports that track compliance KPIs and project alignment with Architecture Principles.
10. **Ensure Continuous Improvement:** Conduct regular meetings and ask for feedback regarding the governance process. Adjust the Architecture Governance Framework based on the feedback received.
- **Hold Quarterly Governance Reviews:** Schedule periodic governance framework reviews with the governance board to assess the effectiveness of governance processes, based on KPIs and feedback.
 - **Gather Feedback:** Regularly solicit feedback from stakeholders, architects, and project teams on how the governance process is working. Use surveys or informal interviews to collect input.
 - **Adjust the Framework:** Based on the feedback and performance data, revise policies, roles, and governance procedures as needed. Document changes and communicate them to the organization.

By following these steps, architecture governance can be implemented effectively. It ensures alignment between the Enterprise Architecture and business objectives, while maintaining control over architectural changes and compliance.



Further Reading

For more information on architecture governance, see “The TOGAF® Leader’s Guide to Establishing and Evolving an EA Capability”, Chapter 6, Architecture Governance, document number G184.

The “Enterprise Architecture Capability and Governance” document of the TOGAF Standard, document number C220, provides additional information about architecture compliance.

8.3. Architecture Board

The Architecture Board is a pivotal component in an organization’s Enterprise Architecture governance, ensuring alignment with strategic goals and fostering architectural best practices.

The Architecture Board – as is shown in Figure 8-1 – is responsible for managing the various architecture roles. This is *functional* rather than *hierarchical* management.

The board, composed of the previously mentioned senior management, key stakeholders, and domain experts, oversees architecture compliance, facilitates decision-making, and manages architecture-related risks. It plays a crucial role in reviewing and approving architecture deliverables, thus ensuring consistency and coherence across the organization. By doing so, the Architecture Board enhances the organization’s ability to adapt and respond to changing business needs effectively.

Contrary to what most people think, an Architecture Board is not a group of architects who meet regularly to discuss substantive issues.

The word *board* refers to the boardroom or board of directors. It is a direct reference to the level at which issues are discussed. Architecture issues discussed at the board level are strategic, not operational.

According to the TOGAF Standard, an Architecture Board is formed by the senior management of an organization. The ideal composition of an Architecture Board consists of C-level management, key stakeholders, and domain experts.

An Architecture Board is a cross-organization entity, which oversees the implementation of the strategy. This body should be representative of all the key stakeholders in the architecture and will typically comprise a group of executives responsible for the review and maintenance of the overall architecture [11].

The reason for this composition is that *strategic decisions* about architecture should be made in an Architecture Board. In organizations, strategic decisions are usually made by C-level management. Therefore, it is necessary to have (some form of) representation from that management at the Architecture Board level.

In practice, however, we often see that an Architecture Board consists of IT Architects. This group of individuals typically deals with substantive issues and topics related to architecture. For example, they deal with architecture designs or discuss (IT) principles and guidelines for a specific solution or solution direction.

Because organizations still view architecture as an IT discipline, in nine out of ten cases an Architecture Board is formed by IT Architects. Due to this misalignment, strategic decisions are not made within the Architecture Board. As a result, these decisions are made outside the eyes and ears of the Architecture Capability. This keeps the Architecture

Capability in a perpetual state of reactive action. It has no say in how the strategic direction is set and implemented. Consequently, architecture remains structurally behind the times.

Examining the Architecture Governance Framework outlined in the TOGAF Standard reveals that an Architecture Board is typically headed by the CIO or CTO – in other words, a C-suite executive (see Figure 8-1).

Personally, I believe that to the two C-level roles that are mentioned in the Architecture Governance Framework should be supplemented with a third, namely, the role of the CEO. When architecture is handled at the CEO level, truly strategic decisions can be made. Architecture is then a direct part of strategy execution.

8.3.1. Diverse Leadership

To illustrate how the governance of the Architecture Board differs when managed by a CIO/CTO or a CEO, two example scenarios are outlined below. The first scenario shows the impact of the Architecture Board being led by the CEO. The second scenario shows how the CIO or CTO would lead the Architecture Board. These two examples illustrate the difference in the strategic impact management can have.

8.3.1.1. Scenario 1: CEO Leadership

This first scenario assumes that the CEO is responsible for the strategic direction of the organization. If, for example, there is a need to ensure that the correct organizational structure, including associated processes, information flows, and applications, is in place due to upcoming legislation, this will have a fairly broad impact on the architecture. Not only will the architecture need to provide an appropriate organizational model,

including the necessary roles and functions, but it will also need to outline a scenario for the processes used and their place within the business function model. In addition, information flows should be mapped and improved according to the strategic direction. Finally, the application landscape should be aligned with the aforementioned organizational design, processes, and information flows.

The scope of the architecture in this scenario is quite broad as it spans multiple architecture domains. The impact of the strategy needs to be mapped across the entire organization.

8.3.1.2. Scenario 2: CIO/CTO Leadership

In the scenario where the Architecture Capability is managed by the CIO or CTO, the scope of architecture work is immediately reduced compared with the scenario where it is managed by the CEO. The more strategic and therefore organization-wide nature of the architecture work is reduced because the CIO or CTO is responsible for a specific part of the strategy. For example, the CIO is only responsible for the information flow, while the CTO is responsible for the technical design and implementation of the architecture.

The situation outlined in the first scenario, where legislation comes into effect, means that in the case of CIO leadership, the architecture work is limited to the design of the information delivery. In the case of the CTO's direction, the architecture work will focus primarily on the technical side of the profession. Thus, both CIO and CTO leadership ensure that the scope of the architecture work is focused on a small part of the overall Enterprise Architecture.

An Architecture Board that is directly led by the CEO - and thus empowered to respond to strategic issues as it should - would look like the one visualized in Figure 8-3. The figure highlights the roles of the CEO, the CIO/CTO, the unit manager, and the Architecture Board itself.

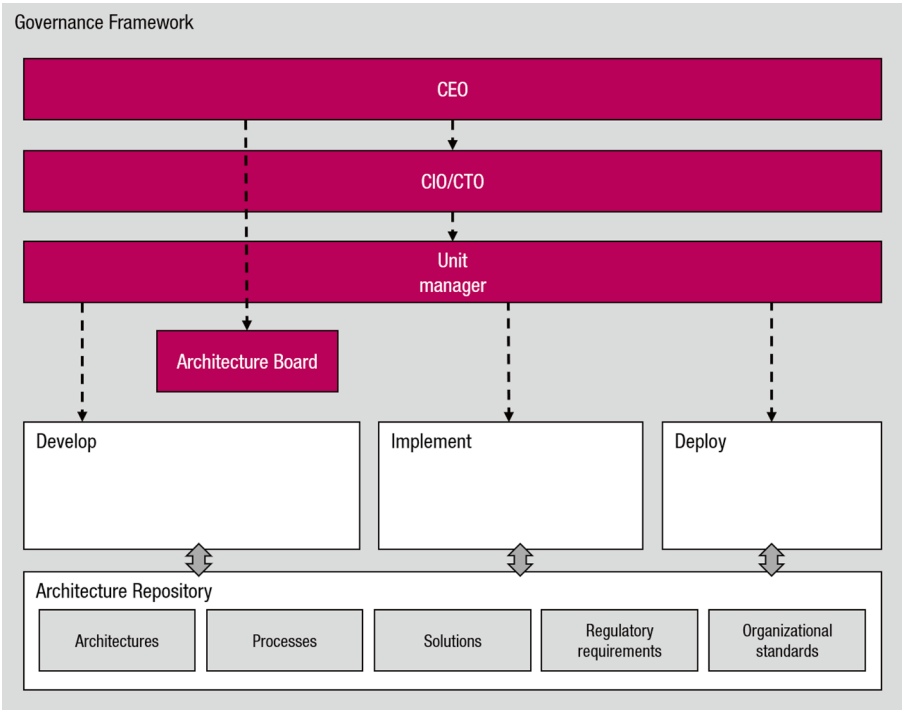


Figure 8-3. *Architecture Board led by the CEO*

In Figure 8-3, the interpretation of the development, implementation, and deployment capabilities has been omitted for the sake of readability. The figure shows that the CEO provides direction to both the CIO/CTO and the Architecture Board. The CIO/CTO is responsible for managing the unit manager(s). The latter role is responsible for the direct management of the various teams involved in the development, implementation, and deployment of the architecture.

As mentioned earlier, strategic decisions are made at the CEO level. By providing direct governance from the CEO to the Architecture Board, it is possible for the Architecture Board to be closely involved in strategic issues. This ensures that the strategic direction of an organization is anchored in the architecture used.

Having an Architecture Board is not just for larger organizations. Smaller organizations in particular can benefit greatly from having an Architecture Board. It is often the smaller organizations that are more pragmatic about rules, so the architecture often has to work harder to maintain certain standards.

Organizations that have a less formal organizational structure can use an Architecture Board. It is by no means necessary for the organization to have roles such as CIO, CTO, or CEO. Companies with a flatter organizational structure can also use an Architecture Board. Architecture sponsorship is critical. If someone high up in the organizational hierarchy can be made responsible for this sponsorship, that person does not necessarily have to be the CEO. What is important is that there is attention and interest from the highest levels of the organization.

8.3.2. When to Set Up an Architecture Board

An Architecture Board is essential for most (somewhat larger) organizations. In many cases, such an Architecture Board already exists, except perhaps for its form and composition. If this is not the case, certain events can trigger the creation of an Architecture Board.

Typically, the creation of an Architecture Board is triggered by one (or more) of the following events:

- The hiring of a new CIO/CTO
- Mergers or acquisitions
- Organizational restructuring, whether or not due to significant business change or rapid growth
- Recognition that the organization's strategy is not being implemented or is being implemented poorly

In many organizations, the CIO (or another C-level executive) is the executive sponsor of architecture efforts. But having a sponsor is not

enough. To gain organization-wide support for the architecture effort, an Architecture Board is needed. This executive-level group is responsible for reviewing and maintaining the strategic architecture.

The Architecture Board is the primary sponsor of the architecture within the organization. But the Architecture Board itself also needs a sponsor, preferably at the highest level within the organization. However, an often-overlooked resource for Architecture Board members is the company's board of directors. Individuals in this role often have diverse knowledge of the company and its competition. Because of their significant influence on the company's vision and goals, they can be successful in validating the alignment of organizational strategy with business objectives. Having the CEO as the Architecture Board's primary sponsor would be an excellent idea.

The involvement of the Architecture Board should extend throughout the planning process and into the maintenance phase of the architecture project.

Unfortunately, many organizations still fail in their attempt to work with architecture. In nine out of ten cases, this is due to a significant lack of involvement and encouragement from senior management.

Performing architectural work depends on solid sponsorship from the highest levels of the organization. Without sponsorship, there is little or no point in doing the work. Therefore, it is important to establish sponsorship and support for architectural work before the work begins.

8.3.3. Key Responsibilities

An Architecture Board has certain responsibilities [11]. Among them are, for example, the following:

- **To Provide a Solid Basis for Decision-Making:**
Decisions regarding the execution of an organization's strategy should never be made lightly. The same goes for decisions regarding architecture, for architecture plays a large and important role in realizing an organization's strategy. Having a solid basis for making these decisions will allow an organization to rely upon the Architecture Board for solid advice.
- **To Ensure Continued Flexibility of the Architecture:**
Every decision made by the Architecture Board must keep in mind that the architecture must do two things in particular. On the one hand, the architecture must always strive to meet the ever-changing needs of the business. On the other hand, an organization should also want to be able to take advantage of new technologies, new ways of working, or new methods. The Architecture Board must ensure that the architecture is flexible enough to achieve these goals, while enforcing architecture compliance.
- **To Improve Architecture Maturity:** The decisions made by the Architecture Board must never allow the architecture in an organization to slip to a lower level of maturity. The Architecture Board must prevent the maturity level from dropping. Decisions on architecture issues should be made with the current and desired architecture maturity levels in mind. Decisions should be aimed at improving the level.

From a more operational point of view, the Architecture Board also holds a number of responsibilities, for example:

- **To Make Sure the Architecture Is Effectively and Consistently Managed and Implemented:** Having a permanent set of members on the Architecture Board ensures a consistent approach to architecture issues. A fixed composition ensures that architecture is not approached in a different way all the time. Of course, the composition of the Architecture Board may change from time to time. One technique is to have fixed terms for members and to have these terms end at different times.
- **To Provide Advice and Guidance:** The Architecture Board has the important operational role of providing advice and direction on all types of architectural issues. Advice must be tested against the existing architecture. The direction given can be translated by experts into concrete actions to implement the architecture work.

Last but not least, the Architecture Board will need to take its responsibilities when it comes to governance. From this perspective, the Architecture Board is responsible for the following:

- **To Provide a Process for Accepting and Approving Architecture Work:** It is critical to establish a process for accepting and approving architectural work. This process should be transparent and accessible to all stakeholders. It is the role of the Architecture Board to provide and make available this process.
- **To Implement a Control Mechanism for Architecture Efforts:** All architecture work and requests for architecture work must at some point pass through a control mechanism to validate that the final

execution of the work is consistent with what was proposed during the development phase. This control usually occurs during Phase G of the Architecture Development Methodology.

- **To Recognize Divergence from the Architecture and Initiatives:** Deviations from the developed architecture may occur over time. Advancing knowledge or other factors play an important role in deviations from the architecture. Identifying and responding to deviations in a timely manner is also a responsibility of the Architecture Board. This is discussed in more detail in Phase H of the Architecture Development Method.

The roles and responsibilities of the Architecture Board should be formally defined and approved by the board's lead sponsor. Ideally, this will be the CEO.

8.3.4. Composition and Size

It is recommended that the Architecture Board consist of four or five people (be careful not to exceed ten). The four or five Architecture Board members are standing members. To keep the Architecture Board at a reasonable size while still ensuring that the entire organization is represented over time, the Architecture Board membership can rotate, giving different senior managers decision-making authority and responsibility. This may be necessary if some Architecture Board members find that a lack of time prevents long-term active participation.

An organization can set up the Architecture Board so that each board member is assigned a number of stakeholders. These stakeholders are then represented at meetings. In day-to-day practice, this means that Architecture Board members need to engage with their stakeholders to get their input on various topics on the agenda.

Rotating individuals within the Architecture Board is quite feasible, provided that care is taken to ensure continuity in the Architecture Board to prevent the Business Architecture from moving from one set of ideas to another. One technique to ensure rotation with continuity is to have fixed terms for members and to have the terms end at different times.

It is also recommended that the composition of the Architecture Board not be exclusively C-level management. The addition of lower-level management and possible experts from the organization, if desired and depending on the issue at hand, can ensure that the tactical and operational aspects of the architecture issue are covered. In this way, the architecture is illuminated and validated from all sides – strategic, tactical, and operational.

On several occasions, I have seen that an Architecture Board made up entirely of C-level executives was not the best solution. This was mainly because the architectural issues at hand were viewed from a too-strategic perspective.

Therefore, it is advisable – depending on the topic and the level of detail of the issue at hand – to use a healthy mix of C-level management combined with lower-level management or experts from the organization as the composition of the Architecture Board. Members drawn from lower management (think unit managers) and experts may join depending on the topic. They are not permanent members of the Architecture Board.

The addition of these additional roles can ensure that issues can be addressed in a more substantive way.

Discussing strategic issues can, in some situations, lead to too much high-level thinking. As a result, the transfer to implementation and execution capabilities may lack the information they need. This makes implementation and execution more difficult.

By involving the above roles (at the request of the Architecture Board), the architecture issue at hand can be considered and addressed from all possible angles. The issue will then receive the strategic, tactical, and even operational attention it deserves.

The size and composition of the Architecture Board may vary depending on the architectural issue at hand. Common sense in the composition of the board is always recommended. In addition, the positioning of the Architecture Capability in the organization plays an important role, as does the way the organization wants to deal with architecture. All of these factors will influence the final composition and size of the Architecture Board. Above all, ensure that the board is able to fulfill its key responsibilities.

8.3.5. How to Set Up an Architecture Board

Implementing an Architecture Board involves setting up a structured *process* and *governance body* that oversees and guides architectural decisions within an organization. This governance ensures alignment with business goals, standards, and best practices.

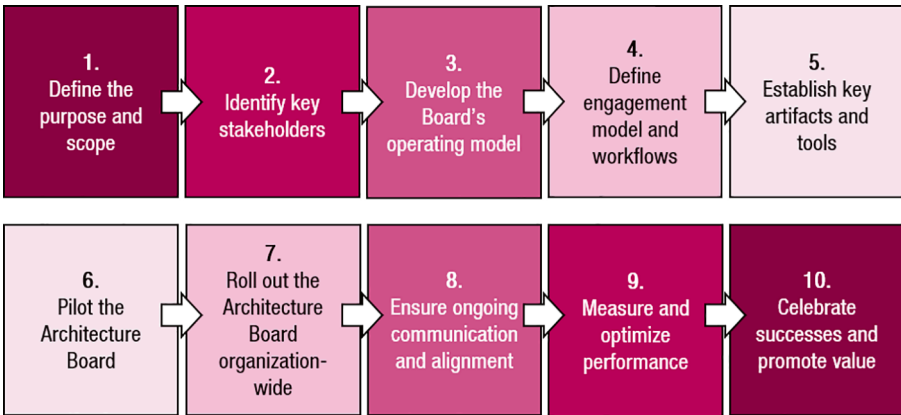


Figure 8-4. *Ten steps for setting up an Architecture Board*

Setting up an Architecture Board in a practical and pragmatic way can be achieved by following the steps shown in Figure 8-4:

1. **Define the Purpose and Scope:** Establish the purpose and boundaries of the Architecture Board.
 - **Identify Objectives:** Determine the key objectives of the Architecture Board, for example, ensuring architectural consistency, approving major architectural decisions, and aligning the architecture with business goals and objectives.
 - **Define Scope:** Clarify the areas of responsibility, such as specific domains (Business, Data, Application, Technology), types of decisions (project approvals, technology selections), and the levels at which the Architecture Board operates (organization-/enterprise-wide, solution specific, or at project level). This level of operations largely depends on the way the Architecture Capability is positioned in the organization.

- **Establish Governance Principles:** Set the guiding principles for decision-making, such as ensuring simplicity, promoting reusability, and adhering to industry standards.
2. **Identify Key Stakeholders:** Ensure representation from relevant stakeholders across the organization.
- **Select Board Members:** Include representatives from key areas such as
 - **Business Stakeholders:** To ensure alignment with business objectives, it is imperative to have business representation on the Architecture Board.
 - **Enterprise Architects:** To bring in the organization-wide perspective.
 - **Segment Architects:** For specific expertise in areas like business, information, data, application, or technology.
 - **Security and Compliance Representatives:** To ensure adherence to security and regulatory requirements.
 - **IT Leaders:** To represent the overall IT strategy.
 - **Project Managers:** For project-level considerations.
 - **Establish Roles and Responsibilities:** Clearly define roles within the Architecture Board. Think, for example, of a chairperson, secretary, or decision-makers. Make sure to also define the responsibilities of each member.

3. **Develop the Board's Operating Model:** Define how the Architecture Board will function on a day-to-day basis.
 - **Set Meeting Cadence:** Decide on the frequency of meetings (e.g., weekly, biweekly) and the length of each meeting. Consider ad hoc meetings for urgent issues.
 - **Agenda Structure:** Develop a standard agenda template, including sections like
 - Review of previous decisions
 - Discussion of new proposals
 - Review of ongoing initiatives
 - Risk and issue management
 - **Decision-Making Process:** Establish how decisions will be made:
 - **Consensus-Based:** Preferable for collaborative environments.
 - **Majority Voting:** If a clear majority is needed for decisions.
 - **Escalation Process:** Define how disagreements or unresolved issues will be escalated.
4. **Define Engagement Model and Workflows:** Create a process for how the Architecture Board interacts with projects and initiatives.

The Architecture Capability can be used to support portfolio management as well as project management. Therefore, it is advisable to clearly define its scope.

- **Submission Process:** Define how teams submit items for board review, including templates for architectural proposals, impact assessments, and decision logs.
- **Review Criteria:** Set the criteria the Architecture Board will use to assess proposals (alignment with strategic objectives, technical feasibility, Risk Management, cost/benefit analysis).
- **Feedback and Iteration:** Develop a process for providing feedback to teams, including a loop for revising and resubmitting proposals based on feedback from the Architecture Board.
- **Decision Documentation:** Ensure decisions are documented, including the rationale behind them, and stored in a central repository for future reference.

In the recent past, I worked for a government organization that regulated the healthcare industry. The organization was an amalgamation of several companies and had existed in its current form for several years before I started working there.

As professional as they thought they were in the work they were doing, the Architecture Capability left a lot to be desired.

This was largely due to management's view of architecture. They saw it – as, unfortunately, many organizations still do today – as something related to IT.

This was the main reason why the Architecture Board, as it was set up, consisted entirely of technicians. In terms of the topics

on the agenda, they did not get much further than discussing technical issues, and they got pretty deep into the weeds. In fact, the Architecture Board at this organization was akin to a meeting of systems engineers, where fiery discussions about the design of a new backup solution were the highlight of the week.

Slowly but surely, I was able to get the various management teams within the organization to play a significant role in providing input on strategic issues. Little by little, I was able to put these issues on the agenda of a newly created architecture advisory group.

As the various management teams saw in practice that something was actually being done with their input, the will and commitment to these new consultations grew. Senior management participation and input grew steadily, creating a platform where strategic and architectural issues were addressed.

The new consultation was eventually renamed “Architecture Board”, while the existing one took on the name “Architects’ Consultation”.

This way, no one lost a much-needed outlet, so there was virtually no opposition.

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5. **Establish Key Artifacts and Tools:** Ensure that the Architecture Board has the necessary tools and documentation to function effectively.

- **Architecture Principles and Standards:**
Document and maintain a set of Architecture Principles, guidelines, and standards that the board will enforce.

- **Decision Templates:** Create standardized templates for decision requests, impact assessments, and architecture reviews.
 - **Collaboration Tools:** Use a collaboration tool or platform to manage documentation, agendas, and minutes.
 - **Tracking and Reporting Tools:** Implement tools to track decisions, action items, and the status of architectural reviews.
6. **Pilot the Architecture Board:** Test the board's processes and workflows in a controlled environment.
- **Select a Pilot Project:** Choose a low-risk project or initiative to pilot the Architecture Board process.
 - **Run a Full Cycle:** Conduct a full cycle of the Architecture Board process, from submission through decision-making and feedback.
 - **Gather Feedback:** Collect feedback from both board members and the project team to identify any pain points or inefficiencies in the process.
 - **Refine the Process:** Make necessary adjustments to the workflows, tools, and engagement model based on the pilot's outcome.
7. **Roll Out the Architecture Board Organization-Wide:** Scale the Architecture Board to cover all relevant projects and initiatives.

- **Communicate the Process:** Clearly communicate the purpose, processes, and expectations of the Architecture Board to the broader organization. This may involve presentations, documentation, or training sessions.
 - **Onboard Projects:** Begin onboarding new projects and initiatives, ensuring they understand how to engage with the Architecture Board.
 - **Monitor and Adapt:** Continuously monitor the effectiveness of the Architecture Board. Collect metrics such as decision turnaround time, compliance with architectural standards, and feedback from project teams.
 - **Iterate on the Process:** Regularly revisit the Architecture Board's processes and structure, making improvements as necessary to adapt to changing business and technological landscapes.
8. **Ensure Ongoing Communication and Alignment:** Maintain alignment between the Architecture Board, other governance bodies, and the wider organization.
- **Regular Updates:** Provide regular updates to senior management and other stakeholders on the outcomes and value delivered by the Architecture Board.
 - **Cross-Board Collaboration:** If multiple governance boards exist (e.g., security board, innovation board), ensure cross-board communication and collaboration.

- **Continuous Education:** Keep the Architecture Board members informed about new trends, technologies, and methodologies that might impact their decisions.
9. **Measure and Optimize Performance:** Continuously evaluate and improve the effectiveness of the Architecture Board.
- **Key Performance Indicators (KPIs):** Establish KPIs to measure the Architecture Board's effectiveness, such as decision turnaround time, adherence to architectural standards, and business value delivered. This last component, delivering business value, is often forgotten. Too often, people get caught up in the content issues, and the business value proposition takes a back seat.
 - **Feedback Loops:** Implement regular feedback loops from both board members and stakeholders to identify areas for improvement.
 - **Continuous Improvement:** Use the feedback and performance metrics to optimize processes, refine governance models, and enhance the Architecture Board's effectiveness.
10. **Celebrate Successes and Promote Value:** Highlight the achievements of the Architecture Board and ensure continued support from the organization. Keep communicating the added business value of the Architecture Board by recognizing milestones. For instance, publicize the board's successes, such as major architectural achievements, process improvements, or contributions to key business

outcomes. It is also recommendable to promote the Architecture Board's value. This can be done by regularly communicating the value that the Architecture Board brings to the organization, ensuring ongoing buy-in from senior leadership and stakeholders.

By following these ten steps, a well-functioning Architecture Board can be created that provides strategic architectural oversight, aligns with business goals, and drives consistency across the organization.

8.4. Summary

This chapter has covered the topics of architecture governance and the Architecture Board. Architecture governance is intended to control and manage an organization's architecture and the Architecture Capability. The TOGAF Standard has developed a framework that shows the possible structure of architecture governance. This framework was explained in more detail and the capabilities included were highlighted.

Architecture Governance can be set up by following these ten actionable steps:

- **Establish the Architecture Governance Framework:** Ensure buy-in from stakeholders by clearly explaining the boundaries and benefits of Architecture Governance.
- **Define governance roles and responsibilities:** Identify, describe and document necessary roles and responsibilities.
- **Develop governance policies and standards:** Form an interest group to develop policies and standards.

- **Set up an architecture compliance process:** Embed the compliance process in the organization by integrating it in the project lifecycle. Remember to include an exception handling process.
- **Implement an architecture review process:** Implement a formal review process for architecture documents by developing templates and documenting the outcome of the review meetings.
- **Establish Key Performance Indicators (KPIs):** Identify metrics to measure the performance of the governance activities.
- **Integrate Governance into the Architecture Development Method:** Define governance checkpoints and integrate them into the Architecture Development Method phases.
- **Develop a Communication Plan:** Identify the audience that is impacted by the architecture governance. Select, create and use appropriate communication materials.
- **Automate governance processes:** If possible, try to automate (parts of) the governance process such as generating reports and review notifications.
- **Ensure continuous improvement:** Conduct regular meetings and ask for feedback regarding the governance process. Adjust the Architecture Governance Framework based on the feedback received.

In addition to architecture governance, the topic of the Architecture Board was also discussed. An Architecture Board enhances the organization's ability to adapt and respond to changing business needs effectively. The ideal composition of an Architecture Board consists of C-level management, key stakeholders, and domain experts. The reason for this composition is that *strategic decisions* about architecture should be made in an Architecture Board.

An Architecture Board is essential for most organizations. If an Architecture Board does not exist, certain events can trigger its creation. The Architecture Board is the primary sponsor of the architecture within the organization. The board itself also needs a sponsor, preferably at the highest level within the organization. An often-overlooked resource for members is the company's board of directors. It is recommended that the composition of the Architecture Board not be exclusively C-level management.

Setting up an Architecture Board is performed by following ten actionable steps:

- **Define the Purpose and Scope:** Establish the purpose and boundaries of the Architecture Board.
- **Identify Key Stakeholders:** Ensure representation from relevant stakeholders across the organization.
- **Develop the Board's Operating Model:** Define how the Architecture Board will function on a day-to-day basis.
- **Define Engagement Model and Workflows:** Create a process for how the Architecture Board interacts with projects and initiatives.
- **Establish Key Artifacts and Tools:** Ensure that the Architecture Board has the necessary tools and documentation to function effectively.

- **Pilot the Architecture Board:** Test the board's processes and workflows in a controlled environment.
- **Roll Out the Architecture Board Organization-Wide:** Scale the Architecture Board to cover all relevant projects and initiatives.
- **Ensure Ongoing Communication and Alignment:** Maintain alignment between the Architecture Board, other governance bodies, and the wider organization.
- **Measure and Optimize Performance:** Continuously evaluate and improve the effectiveness of the Architecture Board.
- **Celebrate Successes and Promote Value:** Highlight the achievements of the Architecture Board and ensure continued support from the organization.

It is important to keep communicating the added business value of the Architecture Board by recognizing milestones.